

EXPERIMENT No. - 52083/
Date: 03/09**TO DETECT THE HETERO ELEMENT OR FOREIGN ELEMENT PRESENT IN THE GIVEN ORGANIC COMPOUND. (Sample E₂)****APPARATUS REQUIRED**

- | | |
|-----------------|--------------------|
| a) Test tube | b) Testtube holder |
| c) Funnel | d) Mortar & pestle |
| e) Filter paper | f) fusion tube |
| g) Burner | h) glass rod |
| i) | j) |

CHEMICAL REQUIRED

- | | |
|---|--|
| a) A piece of Sodium metal | b) Given Organic compound |
| c) NaOH | d) FeSO ₄ |
| e) FeCl ₃ | f) (CH ₃ COO) ₂ Pb |
| g) Na ₂ [Fe(CN) ₅ NO] | h) HCl |
| i) CH ₃ COOH | j) AgNO ₃ |

THEORY

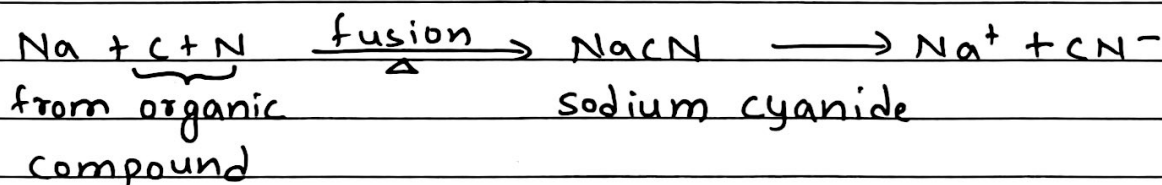
Hydrocarbon & their derivatives in which covalently bonded carbon is essential element are called organic compounds. For eg: ethanol (C₂H₅OH), urea (NH₂CONH₂) etc. All organic compound contain carbon. There are no organic compound without carbon. For eg: CCl₄, C₂Cl₆ etc. Hence parent organic compound are hydrocarbon. Some organic compounds contain other elements, like hydrogen, nitrogen, sulphur, halogen, phosphorus & some metals contain element like oxygen, nitrogen, sulphur, halogen, phosphorus & metals. Hence, elements other

than hydrogen & carbon present in organic compounds are foreign element.

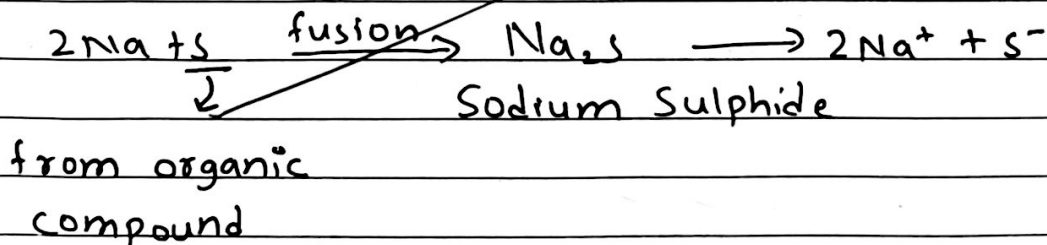
Organic compounds are covalently in nature. Foreign element present in organic compound are also bonded covalently. Therefore the laboratory reagents do not directly react with these compounds. Hence, first these element are converted from covalent to ionic compounds. for this purpose, sodium extract should be prepared.

The sodium fusion test for foreign element was performed by french chemist J.C Lassaigne in 1843 known as Lassaigne's test or Na metal. Following reaction takes place:

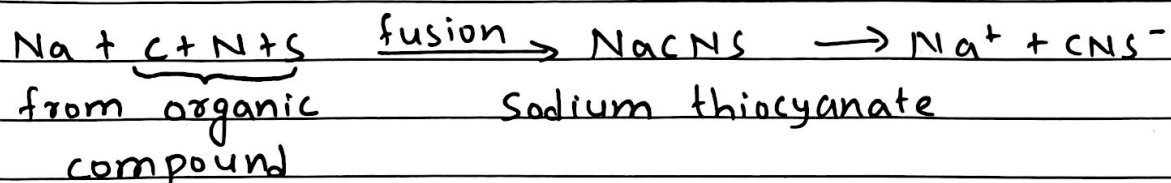
1) If nitrogen is present



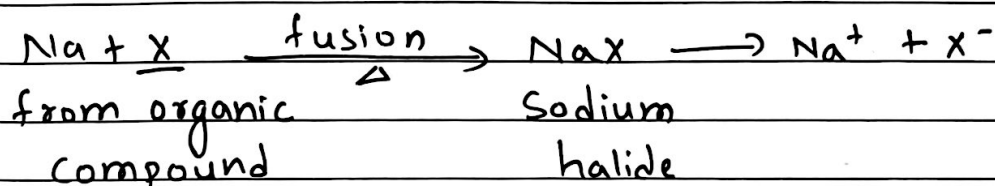
2) If Sulphur is present



3) If both nitrogen & sulphur are present together



4) If halogen is present



PROCEDURE

A freshly cut sodium metal about the size of rice grain was taken. The piece of sodium metal was ~~dried~~ dried by pressing between the folds of filter paper was placed in a clean fusion or ignition tube. The fusion tube was held with the help of tongs and was heated gently till sodium metal fused and formed shining globule. The fusion tube was taken out of the flame & was cooled. Then small amount of organic compound was added. Fusion tube became red hot. The red hot tube was plunged in 10 ml mortar. The tube was broken into small pieces in porcelain basin with the help of pestle. The content was boiled for few minutes and was called sodium extract & this solution was used to perform following tests.

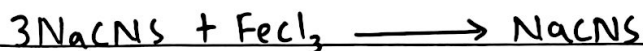
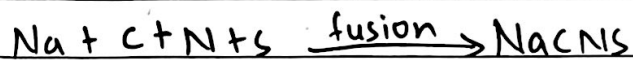
OBSERVATION TABLE

S.N.	Experiments	Observations	Inferences
1.	Detection of nitrogen 1 ml of sodium extract was taken in testtube & few drops of NaOH was added & 1 ml of FeSO_4 sol ⁿ was taken. The mixture was boiled along with shaking & cooled. 1 to 2 drops of FeCl_3 was added & acidified with conc. HCl.	Green colouration was appeared.	presence of nitrogen.
2.	Detection of sulphur a) Sodium nitroprusside test 1 ml of sodium extract was added in clean testtube & few drops of sodium nitroprusside solution was added.	Violet color was appeared on heating	Presence of sulphur.

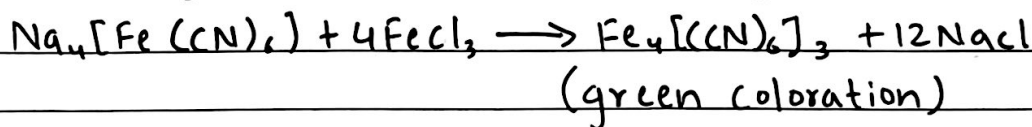
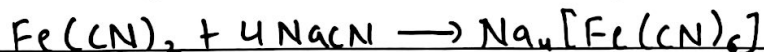
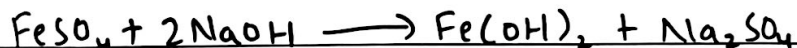
	b) Lead acetate test 1 ml of sodium extract was taken in a clean test tube & few drops of sodium nitroprusside was added.	Black coloration was observed.	presence of sulphur
3.	Detection of both nitrogen and sulphur together 1 ml of sodium extract was taken in a clean test tube & few drops of ferric chloride solution was added.	Blood coloration was appeared	presence of both nitrogen & sulphur.
4.	Detection of halogen 1 ml of sodium extract was taken in a test tube and few drops of conc. HNO_3 was added. The content was boiled, cooled & few drops of AgNO_3 was added.	No ppt was appeared.	Absence of halogen

TEST REACTIONS

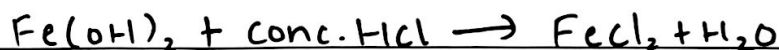
Both sulphur and nitrogen present



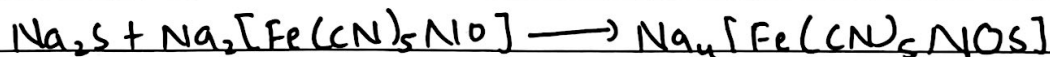
for nitrogen,



For conc. HCl



For sulphur,



Sodium sulphonitroprusside
(Violet colour)



black ppt.

RESULT

In this way, foreign elements present in given organic compound was detected to be nitrogen & sulphur.

CONCLUSION

Hence, from above experiment, it can be concluded that foreign element present in organic compound can be detected using Lassaigne's test.

PRECAUTIONS

- 1) Personal protective equipment should be worn.
- 2) Sodium extract should be prepared in distilled water.
- 3) Apparatus should be handled with care.

VIVA-VOCE

1. What is meant by foreign element?
2. Why is sodium fused with organic compound?
3. Is potassium be used in place of sodium?
4. Why is sodium extract alkaline in nature?
5. Why is it essential to boil sodium extract with nitric acid before halogen test?

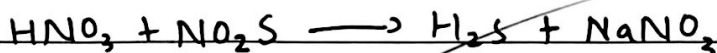
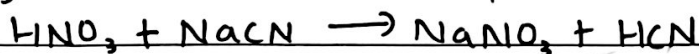
1) Element other than carbon, hydrogen present in organic compound is called foreign element.

2) Foreign elements in organic compound are covalently bonded to carbon so reaction is slow, so they are fused with sodium to convert covalent to ionic.

3) No, potassium cannot be used because it forms poisonous KCN with organic compound and potassium is highly reactive than sodium & reaction is violent.

4) If sodium is not completely reacted with organic compound during fusion, unreacted sodium forms sodium hydroxide with water. Hence it is alkaline in nature.

5) Sodium extract is boiled with nitric acid before halogen test to remove pseudohalide



03/23